Excerpt 1

If you look at a world map or globe, you may be struck by how the coastlines of Africa and Europe (on the eastern side of the Atlantic Ocean) seem to match up with the coastline of the Americas (on the western side). It's not a perfect fit, of course, but the correspondence of these coastlines is striking enough to have fired the imagination of several writers. A number of people before the twentieth century had speculated that these continents might have once been joined together. In 1910, this thought crossed the mind of Alfred Wegener. The major difference between Wegener and the other people who had noticed this fit between the continents is that Wegener looked into the matter more closely. He studied the geology of the African and South American coastal regions, and learned about the plants and animals living there.

Paleontologists had collected a lot of information about the fossils of plants and animals that had once lived on these coasts, and Wegener also studied this fossil record. Much of the information fit together like the pieces of a puzzle, and Wegener became convinced that these continents had once been joined together, 250 million years ago. Since then, they have been drifting apart to their present positions. It turns out that similar correspondences exist between the coasts of east Africa and India.

The idea of continental drift was not accepted by very many people when Wegener proposed it. After all, the thought of continents drifting around like icebergs is rather absurd by common sense standards, and probably aroused opposition for that reason alone. But even on strictly scientific grounds (logic and evidence), there were good reasons not to believe. An alternate explanation for the similarities in geology, fossil record, and plant/animal life had already been advanced and was widely held. This alternate theory assumed that land bridges connected the continents in the past, and these bridges have now sunk under the oceans. Meanwhile, there was a major problem with the drift theory: no mechanism was known that could account for the huge forces that might cause continents to move. Wegener had in fact proposed such a mechanism, but it was easy to show that his proposal must be wrong. He speculated that small differences in gravity between poles and equator might be combined with tidal forces to move the continents. These forces, however, are millions of times too small for this job. Critics of continental drift, ignoring the vast amount of empirical support Wegener had presented, dismissed the theory because his mechanism seemed so clearly wrong. Wegener admitted that his mechanism was speculative, but he insisted that the evidence in favor of moving continents was strong (whatever the mechanism may turn out to be). Sinking land masses (in other words, the proposed former bridges) made no sense to him. Land masses are higher (continents) because they are less dense or lower (ocean floors) because they are more dense. Large land mass areas wouldn't sink or rise at random. Also, certain life forms were found only in a narrow range near both coasts. Surely detached and drifting continents explained this fact better than a vast former land bridge. Finally, Wegener's theory explained why the climates of the continents had been so very different in the distant past (namely, these continents had been located at very different places on the earth's surface then). The opponents of drift also had some good arguments. For example, the lower crust of the earth is not fluid enough to allow the upper crust to move. In their judgement, the geological and paleontological evidence was too fragmentary and incomplete to prove the case for drift. These opponents also charged that many of Wegener's coastal fits were not as good as he claimed (the opponents were mistaken in this case; Wegener had quite rightly used the boundaries of the continental shelf rather than the coastline itself).

Gregory N Derry. What science is and how it works. Princeton University Press, 1999.

Excerpt 2

The branch of philosophy concerned with problems of knowledge is called epistemology. How do we acquire knowledge of the world? What does it mean to know something? How do we know that our knowledge is true? How is our knowledge of something related to the thing itself? These are the kinds of questions explored by the philosophy of knowledge, epistemology. Epistemological questions are broader than the questions found in the philosophy of science, but are in a sense foundational. The roots of the philosophy of science are deeply embedded in epistemology.

Gregory N Derry. What science is and how it works. Princeton University Press, 1999